

Performance Testing & Results

CrediShield AI — Credit Card Approval Prediction System

1. Test Environment

Testing was conducted across both local development environments and the live cloud production deployment to ensure consistency.

Parameter	Specification
Local OS	Windows 11
Local Hardware	Intel Core i5, 8GB RAM
Python Runtime	Version 3.12
Web Server	Gunicorn v22.0.0 (Production) / Flask Dev Server (Local)
Cloud Platform	Render.com (Free Tier: 512MB RAM)
Browsers Tested	Microsoft Edge (v120+), Google Chrome (v120+)

2. Machine Learning Model Performance

The core prediction engine was evaluated on a stratified 20% holdout test set comprising previously unseen application data.

2.1 Model Comparison Matrix

The training pipeline evaluated 6 distinct algorithms. Performance metrics are detailed below:

Algorithm	Accuracy	Precision	Recall	F1-Score	ROC AUC
Logistic Regression	91.45%	96.12%	94.95%	95.53%	93.11%
Decision Tree	95.73%	97.40%	97.98%	97.69%	95.56%
Random Forest	97.85%	98.90%	98.82%	98.86%	96.98%
Gradient Boosting	98.29%	99.21%	99.07%	99.14%	97.53%
XGBoost	97.91%	99.05%	98.77%	98.91%	96.50%
Balanced Random Forest	86.12%	85.50%	88.41%	86.93%	92.84%

2.2 Performance Analysis

- Winner:** Gradient Boosting Classifier.
- Why F1-Score Matters:** Due to severe class imbalance (>98% safe clients), accuracy is misleading. The F1-score of 99.14% confirms the model achieves an outstanding balance between Precision (not approving risky clients) and Recall (not rejecting safe clients).

3. System Functional Testing

Test Case 1: Safe Profile Validation (Approval Scenario)

Parameter	Detail
Objective	Verify the model correctly identifies a low-risk applicant.
Input Vector	Male, 30 yrs, \$120k income, Higher Education, Married, 1 child, House/Apt, Manager, 5 yrs employed, Family size 3, Owns Car, Owns Realty, Working
Expected Result	Approved (True) with High Confidence
Actual API Output	<code>{"approved": true, "confidence": 98.76, "risk_level": "Low"}</code>
Status	☒ PASS

Test Case 2: Risky Profile Validation (Rejection Scenario)

Parameter	Detail
Objective	Verify the model correctly identifies a high-risk/delinquent applicant.
Input Vector	Female, 22 yrs, \$15k income, Lower Secondary, Single, 0 children, Renting, Laborer, 0 yrs employed (Unemployed), Family size 1, No Car, No Realty
Expected Result	Rejected (False) with High Risk
Actual API Output	<code>{"approved": false, "confidence": 87.34, "risk_level": "High"}</code>
Status	☒ PASS

Test Case 3: Invalid Input Handling

Parameter	Detail
Objective	Ensure the API does not crash when required fields are missing.
Input Vector	JSON payload omitting the <code>gender</code> and <code>income</code> fields.
Expected Result	HTTP 400 Bad Request with a descriptive error.
Actual API Output	<code>{"error": "Missing input fields: gender, income"}</code> (HTTP 400)
Status	☒ PASS

4. Non-Functional Performance Testing

4.1 Latency and Response Time

Metric	Threshold	Target	Actual Measured	Status
API Prediction Inference Time	< 500 ms		~85 ms (average)	☒ PASS
UI Initial Load Time (Render)	< 3.0 seconds		~1.2 seconds	☒ PASS

(Note: Response times were measured under standard load conditions. Render free-tier instances experience a known ~30-second "cold start" delay if inactive for 15 minutes, after which response times return to <100ms).

4.2 Browser & UI Compatibility

The glassmorphism UI and Chart.js rendering were tested manually across viewport sizes:

- **Desktop (1080p):** Rendered perfectly. Animations smooth.
- **Tablet (768px width):** CSS Flexbox grids adjusted correctly. Navigation remained accessible.
- **Print Layout:** The hidden iframe triggered the PDF/Print dialog successfully, and the CSS @media print rules stripped unnecessary UI elements to generate a clean "Assessment Letter."

4.3 Deployment Stability

- **Gunicorn Port Binding:** Verified that the application binds correctly to 0.0.0.0 and utilizes the Render-injected PORT environment variable.
- **Dependency Deserialization:** Verified that pinning scikit-learn==1.5.0 resolved the _loss module ModuleNotFoundError during cloud deployment model unpickling.

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Date: July 2026